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To the Editors, *SIAM Journal on Mathematics of Data Science*

Dear Editors,

Please consider the enclosed manuscript, “Toward the Angular-Preservation Theorem: Reductions, a Non-Radiality Calculus, and the Green-Rank Frontier,” as a companion to my submission “Keep the Angle: A Geometry-Preserving Basis in Spectral Embeddings” (Paper I). I am writing in the first instance to give the editors the *context* in which this follow-up sits relative to that paper, and to make its relationship to Paper I fully transparent.

The relationship to Paper I. Paper I proves that the geometry a large graph Laplacian appears to lose to the commute-time collapse survives in the *angular* coordinate of the spectral embedding, and it isolates one central open statement—the rank form of *Angular Preservation*—which it reduces, for intrinsic dimension $d \leq 3$, to two ingredients: a graph-to-continuum rank transfer and *Green-rank positivity* ($\rho_G > 0$, a positivity property of the Green kernel’s rank correlation with distance). Paper I states these as a conjecture and a conjecture, respectively, and explicitly does not attempt them. The present manuscript takes up exactly those two open items and moves them toward theorems. It therefore *depends on* Paper I for its framework, definitions, and the Green-kernel rank-limit theorem it builds upon, and it *shares no theorem* with Paper I: where Paper I conjectures, this paper proves, deletes hypotheses, or reduces to an explicit geometric quantity. The two are a paper and its theoretical sequel, not overlapping submissions.

What the follow-up establishes. Briefly: (i) the auxiliary hypotheses (H3) and atomlessness of Paper I’s conditional results are deleted outright; (ii) the transfer’s anti-concentration hypothesis, false in the bounded-density form stated informally in the program’s notes, is proved in the operative *modulus* form for every real-analytic metric; (iii) a non-radiality calculus reduces Green-rank positivity to a single *scale-invariant* index of the Green kernel; (iv) that index is bounded geometrically—it is the within-shell variance of the heat kernel, vanishing on two-point homogeneous spaces, finite exactly for $d \leq 3$, with a quantitative $\eta^{2/3}$ perturbative openness rate—leaving one named curvature-anisotropy amplitude as the sole non-explicit constant; and (v) the whole chain is carried past the critical dimension four by a fractional filter, with the exact symmetric-space anchor proved in every dimension. Green-rank positivity is thereby established on an increasing hierarchy of explicit classes, and reduced in general to one geometric coefficient; a numerical counterexample hunt across four geometry families (in the public repository) finds none.

On timing and how the editors might use this. I recognize that a follow-up is ordinarily considered after the primary paper; I am submitting the context now so that the relationship is on the record and the editors may decide how to proceed—whether to hold this manuscript pending Paper I’s disposition, to review it in parallel, or to treat it as a standalone contribution to the Green-kernel/spectral-embedding literature (it is written to be read alongside Paper I but is self-contained at the level of statements). I defer entirely to the editors’ preference.

Originality and exclusivity. This manuscript is original, has not been published, and is not under review at any other journal or conference. Its only overlap with prior work is its dependence on Paper I, disclosed above; it shares no theorem with any other submission. I intend to post it as a preprint, consistent with SIAM policy.

Preparation, AI assistance, and status, stated candidly. I want to be forthright about how this manuscript was produced. It was developed with substantial assistance from an AI system, and its internal checking was carried out by fresh-context adversarial re-derivations run on the same class of system; that process caught and corrected numerous errors in preparation (a parametrix integrability condition, an optimization step, a curvature coefficient, among others). A further round of *AI-assisted* expert-style review has since been run (again a large language model, prompted adversarially), and its corrections are incorporated—including the retraction of a false $d = 4$ “coincidence” and repairs to the perturbative Duhamel argument (now the honest $t^{-d/4}$ envelope, valid for $d \leq 3$) and the subordination argument (now grounded in a published sharp heat-kernel estimate). *No human expert has yet refereed the manuscript.* I state all this plainly because AI-assisted verification is not a substitute for refereeing or machine-checked proof; the elementary core has in addition been checked by computer algebra. The surviving core (the master and distributional inequalities, the anti-concentration modulus result, and the fractional Riesz observation) is, I believe, a real advance; but the results are largely *explicit-class* statements reducing the conjecture rather than resolving it, and the differential geometry of Parts IV–V in particular warrants an expert referee’s scrutiny. I own every claim in the paper and have tried to carry the open problems and honest limits into the text rather than around it.

Thank you for your consideration, and for allowing me to place this work in its proper relation to the paper it follows.

Sincerely,
Andrew H. Bond